Chapters 1-5

Frequency Interpretation - repeated large number of times under similar conditions.

Classical Interpretation - concept of n, equally likely outcomes.

Subjective Interpretation - based on a person's assignment of judgement for a certain outcome.

Set Theory Exercises)

1. Suppose $A \subset B$. Show that $B^c \subset A^c$.

We must prove that $B^c \subset A^c$. Let there be an element $x \in B^c$.

Then, by definition of compliments, $x \notin B$. Since $A \subset B$, the contrapositive gives us: if $x \in B$, then $x \in A$.

Since contrapositive $\equiv$ original statement, $x \in A^c$: and since x is arbitrary, $B^c \subset A^c$.

14. Suppose that the sample space S of some experiment is countable. Suppose also that, for every outcome $s \in S$, the subset $\{s\}$ is an event. Show that every subset of S must be an event.

Let A be any subset of S. S is countable, so A is countable.

$\{s\}$ is an event $\forall s \in S$.

So $A = \{s_1\} \cup \{s_2\} \cup \{s_3\} \cup \dots$. Each $\{s_i\}$ is an event by assumption. Since a countable union of events is an event, A is an event.

Three Axiom Exercise)

10. For two arbitrary events A and B, prove that

$$Pr(A) = Pr(A \cap B) + Pr(A \cap B^c).$$

Let A and B be arbitrary events. Any element $y \in A$ is either in B or not in B.
Therefore,

$$A = (A \cap B) \cup (A \cap B^c).$$

These two sets are disjoint since no element can be in both B and B^c simultaneously.
By Axiom 3:

$$Pr(A) = Pr(A \cap B) + Pr(A \cap B^c).$$

Probability: assume you have a known probability distribution, you quantify the likelihood of future data/samples through the known.

flip

Statistics: the data/samples are known and you are instead trying to determine the probability distribution.

The probability of an event `A` out of a set of possibilities that can occur.

$$P(A) = \frac{\text{\# of ways A can happen}}{\text{\# of things that can happen}}$$

ex) I flip a coin twice. What is the probability of...

i) A = atleast one heads

$\Omega = \{hh, ht, th, tt\}$ *set omega of all possibilities

$A = \{hh, ht, th\}$

$P(A) = \frac{3}{4}$

ii) B = no heads

$B = \{tt\}$ $P(B) = \frac{1}{4}$

*super simple but doing anyway

So if we scale this up to 10 coin flips it becomes harder to write all possibilities without sequences or powers. The number of possibilities for 10 coin flips is 2^{10}. (order matters and with replacement)

ex)

Deck of 52 cards. How many 5 card runs can I deal off the top of this deck?

$\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }\ \ldots$
52 51 50 49

without replacement

$= \frac{52!}{47!}$

* $n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$

Given n choices and sampling r elements from n, the number of possibilities with replacement is n^r, and $\frac{n!}{(n-r)!}$ BOTH WITH ORDER. If order doesn't matter, say the river in poker, then it would be:

$$\frac{n!}{(n-r)!\, r!} = \binom{n}{r} = nCr$$

Practice Problems)

1) a PIN code is 4 digits (0-9). How many possible PIN's exist if digits can repeat? If they can't?

i) $\underset{0\text{-}9}{\underline{\ }}\ \underset{0\text{-}9}{\underline{\ }}\ \underset{0\text{-}9}{\underline{\ }}\ \underset{0\text{-}9}{\underline{\ }} = 10^4$ (10 choices for each 4 slots) = 10,000

ii) $\frac{10!}{(10-4)!} = \frac{10!}{6!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot \cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}}{\cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}} = 10 \cdot 9 \cdot 8 \cdot 7 = 5{,}040$

2) a license plate has 3 letters followed by 3 digits. Letters and digits can repeat. How many plates are possible?

$$\underline{\quad}_{27}\ \underline{\quad}_{27}\ \underline{\quad}_{27}\ \underline{\quad}_{10}\ \underline{\quad}_{10}\ \underline{\quad}_{10} = 27^3 \cdot 10^3 = 19683 \cdot 1000$$

$= 19{,}683{,}000$ possible license plates

3) a class of 30 students. How many ways can you choose a president, vice president, and treasurer?

$n = 30$ students

$r = 3$ possible roles; each student eligiable and w/o replacement

$$\frac{n!}{(n-r)!} = \frac{30!}{(30-3)!} = \frac{30!}{27!} = 30 \cdot 29 \cdot 28 = 24360 \text{ ways}$$

4) a poker hand is 5 cards from a 52 card deck. How many distinct hands are possible?

$n = 52$

$r = 5$; w/o replacement & no order

$$\frac{n!}{(n-r)! \cdot r!} = \frac{52!}{(52-5)! \cdot 5!} = \frac{52!}{47! \cdot 5!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{311875200}{120}$$

$= 2598960$ distint hands possible

5) we have 12 books. How many ways can you arrange any 4 of them on a shelf? How many ways can you choose 4 of them?

$n = 12$

$r = 4$; w/o replacement

no order

$$\frac{12!}{(12-4)! \cdot 4!} = \frac{12!}{8! \cdot 4!} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1}$$

$= 495$ ways to chose 4 books

with order

$$\frac{12!}{(12-4)!} = \frac{12!}{8!} = 12 \cdot 11 \cdot 10 \cdot 9$$

$= 11{,}880$ ways to arrange 4 on a shelf

Axioms and Basic Theorems

an axiom is a statement or proposition which is regarded as being established, accepted, or self-evidently true.
*the axiom is that supply equals demand.

Axiom 1 For every event A, $Pr(A) \geq 0$.

The second axiom says that if an event is certain to occur, then the probability of that event is 1.

Axiom 2 $Pr(S) = 1$.

Axiom 3 For every infinite sequence of disjoint events $A_1, A_2, A_3, \ldots$

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i).$$

Probability. A probability measure, or simply a probability, on a sample space S is a specification of numbers $Pr(A)$ for all events A that satisfy Axioms 1-3.

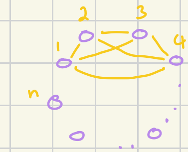
Consider the infinite sequence of events $A_1, A_2, \ldots$ such that $A_i \neq \emptyset$ for $i = 1, 2, \ldots$. In other words, each event in the sequence is the empty set $\emptyset$. Then this sequence is a sequence of disjoint events, since $\emptyset \cap \emptyset = \emptyset$. Furthermore, $\bigcup_{i=1}^{\infty} A_i = \emptyset$. Therefore, it follows from Axiom 3 that

$$Pr(\emptyset) = Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i) = \sum_{i=1}^{\infty} Pr(\emptyset).$$

This equation states that when $Pr(\emptyset)$ is added repeatedly in an inf. series, the sum of that series is simply $Pr(\emptyset)$.

The Birthday Problem. If there are n people in a room. How large does n need to be for there to be a 50% chance that 2 people have the same birthday?

* easiest way is to find the inverse (no birthday) and subtracting from 1



$$(n-1)+(n-2)+(n-3)+\dots+2+1 = \frac{n(n-1)}{2}$$

$$P(\geq 2 \text{ share}) = \underbrace{1 - P(\text{no share})}_{\text{way easier to compute}}$$

$$P(\geq 2 \text{ share}) = P(\text{exactly } 2) + P(\text{exactly } 3) + \dots$$

very long

$$\underbrace{\frac{365}{365}}_{1} \; \underbrace{\frac{364}{365}}_{2} \; \underbrace{\frac{363}{365}}_{3} \; \dots \; \underbrace{\frac{365-n+1}{365}}_{n}$$

$$P(\text{no-one shares}) = \frac{\#\text{ of unique birthdays}}{\underbrace{\#\text{ choices}}_{365^n}}$$

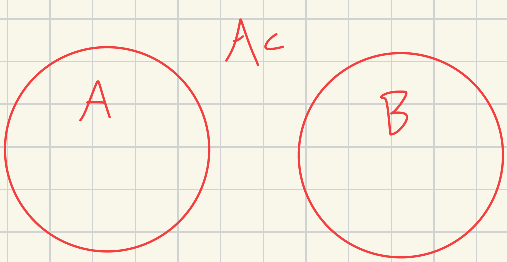
$$\frac{365!}{(365-n)! \cdot 365^n}$$

Practice Problem)

1) consider two events A and B such that $Pr(A) = 1/3$ and $Pr(B) = 1/2$. Determine the value of $Pr(B \cap A^c)$ for each of the following conditions:

a) A and B are disjoint.

$Pr(A) = 1/3$
$Pr(B) = 1/2$

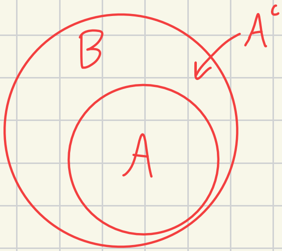


Since A^c is the whole space and all of B already lives outside of A:

$B \cap A^c = B \rightarrow \boxed{Pr(B) = 1/2}$

b) $A \subset B$.

$Pr(A) = 1/3$
$Pr(B) = 1/2$



so $B \cap A^c = B - A$

$B - A = \frac{1}{2} - \frac{1}{3}$ or $\frac{3}{6} - \frac{2}{6} = \frac{1}{6}$

$\boxed{Pr(B-A) = 1/6}$

a random variable is a function that maps each outcome in a space to a number.

Coin flip : $X(\text{heads}) = 1$, $X(\text{tails}) = 0$

Die roll : $Y(\text{face}_i) = i$.

* the randomness lives in which outcome happens and the variable is just numerical encoding.

Variable is either discrete (on a countable set i.e $\mathbb{Z}, \{1, 2, \dots\}$), or continuous (i.e. $\mathbb{R}$).

discrete

a probability mass function (PMF) is a function $p(k)$ giving $P(X = k)$ for each possible value k.

i) $p(k) \geq 0$ for every k

ii) $\sum p(k) = 1$, summed over all possible k

iii) $P(X \in S) = \sum$ over $k \in S$ of $p(k)$, for any set S of values continuous

a probability density function (PDF) come from integrating and not summing.

* will return later

Bernoulli random variable takes 1 with $P(p)$, and 0 with $P(1-p)$.

$$p(k;p) = p^k \cdot (1-p)^{1-k}, \text{ for } k \in \{0,1\}$$

plug in $k=1$: $p^1 \cdot (1-p)^{1-1} = p^1 \cdot (1-p)^0 = p$ ✓

plug in $k=0$: $p^0 \cdot (1-p)^{1-0} = p^0 \cdot (1-p)^1 = 1-p$ ✓

ex)

Three independent coin flips, each Bernoulli$(p=0.6)$. Whats the probability of observing the sequence $(1,0,1)$?

all possabilities = $\{(1,1,1), (1,1,0), (1,0,0), (0,1,1), (0,0,1), (0,0,0), (1,0,1), (0,1,0)\}$

coin is not fair ($p=0.6$ not 0.5) so 60% of 1.

$P(1,0,1) = 0.6 \cdot 0.4 \cdot 0.6 = \underline{0.144}$

A student selected from a class will either be a boy or a girl. If the probability that a boy will be selected is 0.3, what is the probability a girl will be selected?

Let B represent the set of boys and G represent the set of girls. The classroom will therefore represent the sample space S. Using the information given:

$$Pr(B) = 0.3, \quad Pr(S) = 1 \text{ by axiom 2}$$

Since B and G are disjoint, the $Pr(G)$ is ...

$$Pr(G) = Pr(S) - Pr(B) \rightarrow Pr(G) = 1 - 0.3 = \underline{0.7}$$

If two balanced dice are rolled, what is the probability that that the sum of the two numbers will be odd?

balanced so equally as likely

Let A represent the space of all possible values of die one and B represent all possible values of die 2. That means there exists an $a \in A$ and $b \in B$ such that:

$$A = \{1, 2, 3, 4, 5, 6\}$$
$$B = \{1, 2, 3, 4, 5, 6\}$$

and the sum is odd if one dice is odd and one is even.

$$a + b = 2k+1$$
$$2k + 2k+1 = 2k+1 \checkmark$$
$$2k+1 + 2k = 2k+1 \checkmark$$

each roll is $1/36$ since the rolling of 2 dice is 36 possible outcomes. If one must be odd and other even, we can count all the possibilities to be $18/36$ or $1/2$. Same for even since its just $B^c + A^c$.

If 3 fair coins are tossed, what is the probability that all 3 faces will be the same?

$x = h, t$

$y = h, t$ $\quad (x, y, z) = 2 \cdot 2\,2 = 2^3 = 8$

$z = h, t$

each $1/8$. Since HHH, TTT only two where all are same $1/8 + 1/8 = 2/8$ or $1/4$.

Consider an experiment which a fair coin and a fair die are thrown.

a) what is the sample space?

x = coin
y = dice

$(x, y) = (h,1), (h,2), (h,3), (h,4), (h,5), (h,6), (t,1), (t,2), (t,3), (t,4), (t,5), (t,6)\}$

$2 \cdot 6 = 12 \quad 1/12$ each

b) what is probability of head + odd die?

prob of head is $1/2$ since heads/tails

odd is also $1/2$. but together only make up $3/12$ or $1/4$.

Combinatorial Methods

Consider a set $\{a, b, c, d\}$, the number of distint subsets of size 2 is:

$$\{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\}, \text{ and } \{c,d\}.$$

* this is countable, but now imagine a larger set of either subsets or permientations. This would be impossible to count

So... consider a set with n elements. Each subset of size k chosen from this set is called a combination of n elements taken k at a time. Which is denoted as:

$$C_{n,k}$$

and when looking at a permientation, rather than a subset, there are $k!$ ways to arrange those k elements out of n.

$$C_{n,k} = \frac{P_{n,k}}{k!} = \frac{n!}{k!(n-k)!}$$

$$n = \{\overset{1}{a}, \overset{2}{b}, \overset{3}{c}, \overset{4}{d}\}$$
$$k = \{\overset{1}{a}, \overset{2}{b}\}, \ldots$$

$$C_{4,2} = \frac{P_{4,2}}{2!} = \frac{\cancel{4!}}{\cancel{2!}2!} = \frac{4 \cdot 3}{2 \cdot 1} = \frac{12}{2} = 6$$

ex) Suppose that a committee of 8 people is to be selected from a group of 20, and each of the 8 members get a different job to perform.

the number of ways to chose 8 out of 20 and assign them different jobs is the number of permutations of 20 elements taken 8 at a time, or...

$$P_{20,8} = C_{20,8} \cdot 8! = 125,970 \cdot 8! = 5,078,110,400$$

$C_{n,k}$ is written as ...

$\binom{n}{k}$, for

$k = 0, 1, \ldots, n$

$$\underbrace{\binom{n}{k}}_{} = \frac{n!}{k!(n-k)!}$$ called a binomial coefficient

- "the number of combinations of n elements taken k at a time"
- "the binomial coefficient of n and k"
- "n choose k"

$$\frac{n(n-1)(n-2)\ldots(n-k+1)}{k!} = \binom{n}{k}$$

$$\binom{10}{3} = C_{10,3} \text{ or } 10C_3 = \frac{10 \cdot \overset{10-1}{9} \cdot \overset{10-3+1}{8}}{\underset{3!}{3 \cdot 2}} = 120$$

ex)

Suppose that n people are seated in a random manner in a row of n theater seats. What is the probability that two particular people A and B sit next to eachother?

$$p = \frac{n-1}{\binom{n}{2}} = \frac{n-1}{\frac{n(n-1)}{2}} = \frac{2(n-1)}{n(n-1)} = \frac{2}{n}$$

Working through problems:

factorial $n = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 1$

$12 \cdot 12-1 \cdot 12-2 \cdot \ldots \cdot 1$

from $1 \rightarrow n+1$

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